Cura :
AI-Powered Personalized Reminder System for Diet and Medicine

IBM AI-ML INTERNSHIP PROJECT

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Introduction

- In today's fast-paced lifestyle, maintaining a consistent diet and medication schedule is a challenge.

- **Cura** is a smart AI-powered reminder system that sends personalized alerts for:
  - ✓ Medicine intake
  - ✓ Meal timings

- Adapts to user routines, preferences, and health conditions.

- Promotes wellness via consistency and personalization.

Official
Problem Statement

"Ensure healthy lives and promote well-being for all at all ages"

Improved
Problem Statement

- Elderly or busy individuals often forget medication or follow generic, ineffective diets.

- Existing reminder tools (like alarms or static apps) are:
  - Not personalized
  - Not context-aware
  - Ineffective for long-term adherence

- There is a critical need for an **adaptive and smart health reminder system**.

Project Objectives

1. Create a time-based medicine reminder system.
2. Provide **goal-oriented diet recommendations** (e.g., diabetes, weight loss).
3. Design a simple, **user-friendly interface**.
4. Increase adherence through smart, adaptive notifications.

Why This Problem Matters?

- Missed medicines or poor diets → chronic issues, hospitalizations.

- Studies show low adherence rates among:
  - ✓ Elderly
  - ✓ Working professionals

- Generic notifications = ignored

- AI makes it possible to **learn, adapt, and personalize**.
- Cura aims to bridge the **gap in personalized health management**.

Proposed Solution

- **Cura = AI-powered health companion**.
- Two main features :

1

Medicine Reminder Agent

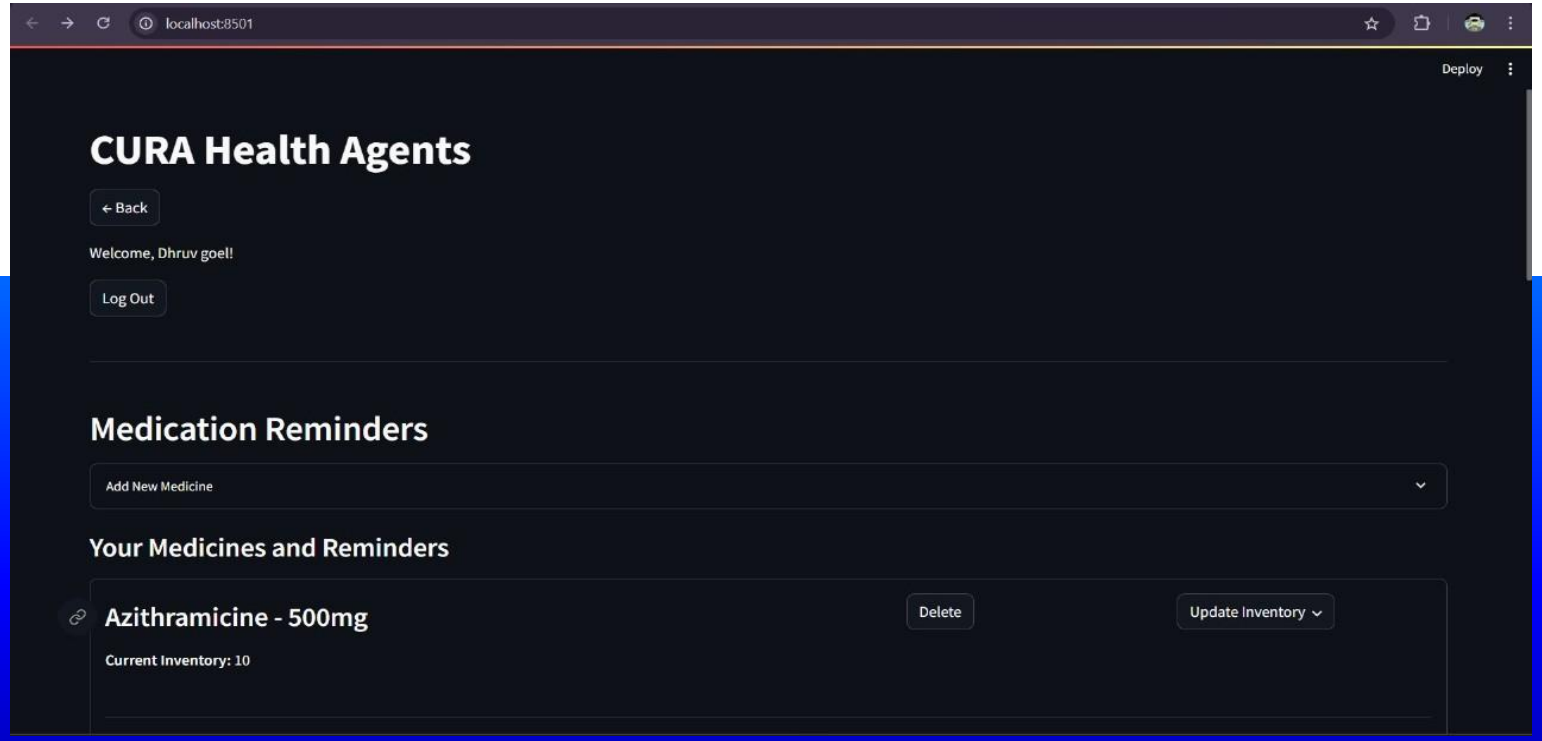
- Smart, adaptive alerts for medicine intake.
- Schedule and receive notifications for their medications.

2

Diet Reminder System

- Adapts to user feedback and evolving goals for continuous improvement.
- Scheduled meal alerts.

Cura
Interface Overview



“Cura’s intuitive interface for managing health reminders.”

Key Features

- Adaptive AI logic
- Track stock and refill thresholds
- Smart medicine and meal scheduling
- Simple, intuitive user interface
- Personalized email alerts
- Built for ML model integration in the future

Technologies
Powering CURA

- **Python** – Core programming language for all services
- **Streamlit** – Interactive, lightweight front-end interface
- **Django REST Framework (DRF)** – Backend API handling & orchestration
- **Flask** – Microservice for reminder agent
- **PostgreSQL** – Reliable, scalable database solution

Development
& Deployment Tools

- **Git & GitHub** – Version control & collaboration
- **Poetry** – Dependency & environment management
- **Render** – Cloud deployment for backend microservices

Core Python Dependencies

- Front-End (Streamlit)
  - Streamlit
- Django Backend
  - Django>=4.0
  - Requests
  - langchain-google-genai
  - Pydantic
  - python-dotenv
  - django-cron
- Reminder Agent (Flask Service)
  - Flask
  - SQLAlchemy

Source Code Links

- **Backend:** github.com/abhay-byte/cura-backend

- **Frontend:** github.com/spakigamer/CURA_frontend

System Architecture

- **Type:** Microservices architecture
- **Structure:**
  - **Central Django backend** orchestrates AI agents
  - **Separate Flask reminder service** for notifications
  - **Streamlit frontend** communicates with backend APIs
  - **PostgreSQL database** stores user, diet, and reminder data

Try CURA
in Action

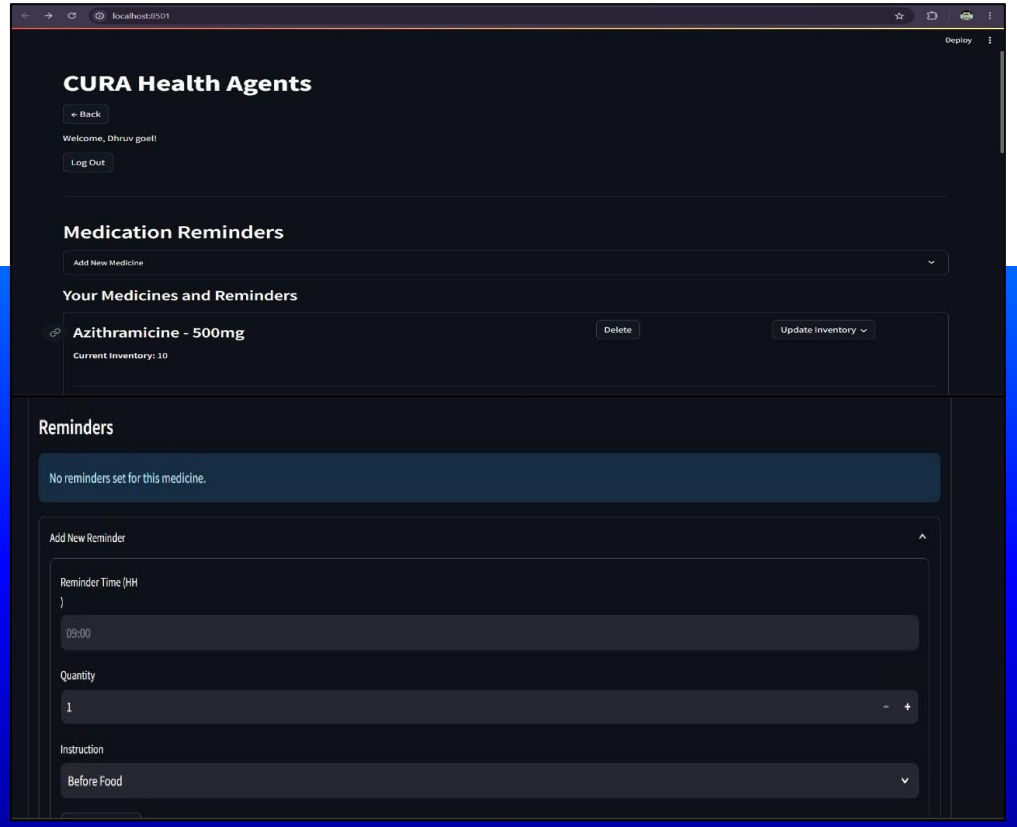
- **Frontend (Streamlit):**

curaproject.streamlit.app

- **Backend API (Django REST):**

cura-backend-main-99c8.onrender.com

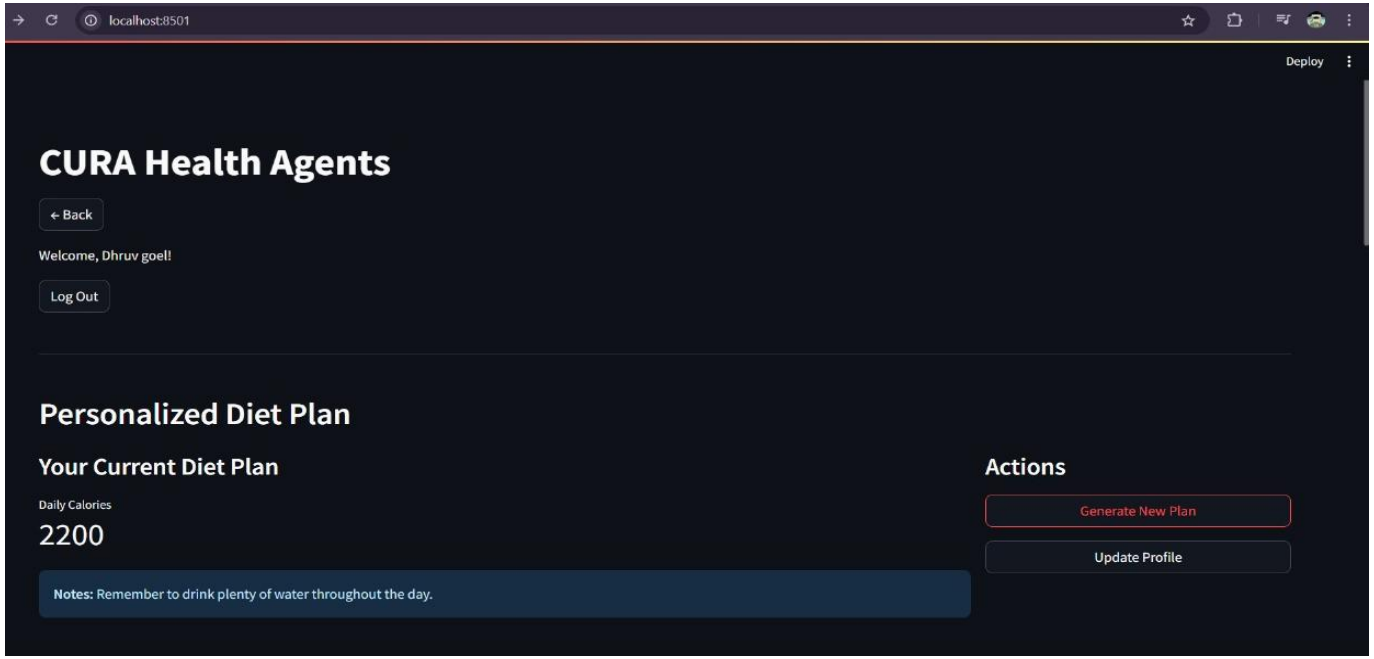
Feature Walkthrough :
Medication Reminder Setup



CURA Web Interface :
Health Profile Input

"Users enter personal health details including age, weight, activity level, and goals. This data powers Cura's personalized diet and medication recommendations."

CURA Web Interface : Personalized Diet Plan



"Users receive tailored diet plans based on health goals and preferences. Daily calorie targets and wellness tips enhance adherence."

Implementation Details

Data Handling

- User data: email, medicine, dosage, time, stock, etc.
- Stored securely in SQLite

Logic & Features

- Reminder engine checks for due medicines
- Refill alert system monitors stock

Testing

- Triggered via API endpoints
- Manually validated delivery + stock update

Live Backend &
API Documentation

https://cura-backend-main-99c8.onrender.com

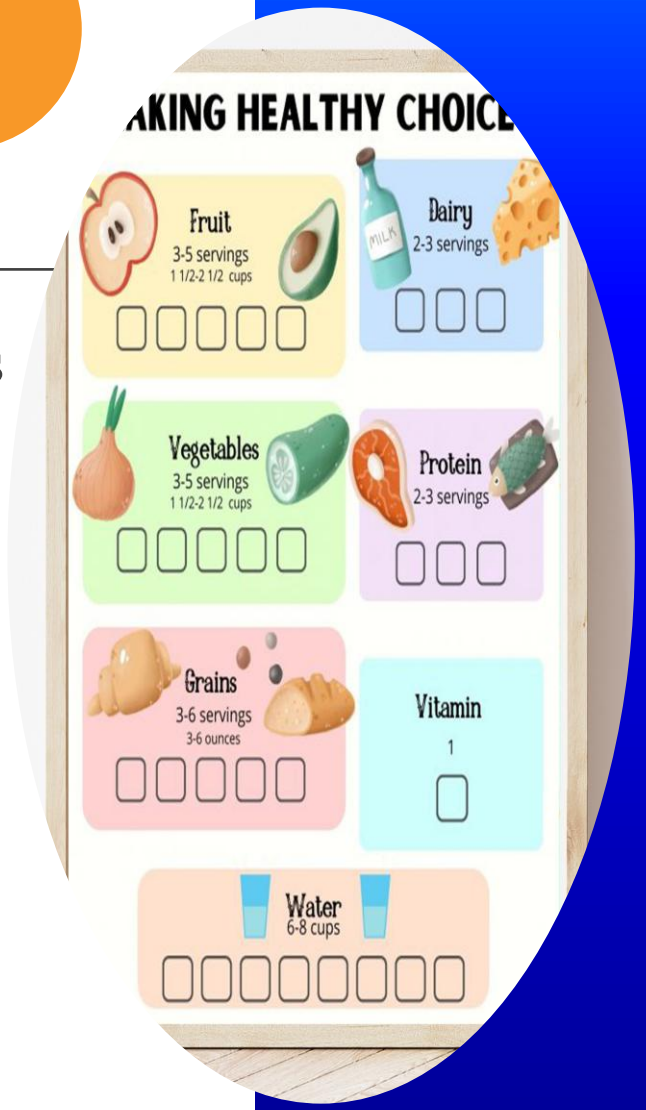
Pretty-print ☑


```
{
  "status": "ok",
  "message": "Welcome to the CURA AI Health Agents API!",
  "documentation": "https://github.com/abhay-byte/cura-backend/blob/main/docs/api.md"
}
```


"Cura's backend API is live and documented. This ensures seamless integration and future scalability."

Planned Enhancements

- **Adaptive Reminders:** Use ML to optimize medication timings
- **Interactive Notifications:** Mark doses as *taken* or *snooze* directly
- **Grocery List Integration:** Auto-fill carts on Blinkit/Zepto from meal plans
- **Weekly Diet Planning:** Generate 7-day comprehensive plans
- **Nutrient Tracking:** Monitor vitamins, minerals alongside macros



Conclusion

- Cura provides a **smart, practical solution** for health management.
- Enables better **medication and diet adherence** using AI-driven reminders.
- Built with **scalability and future ML integration** in mind.
- Has real-world potential to **improve public health and chronic care outcomes**.

“Thank you for your time!”

We welcome your feedback and questions.